Tutorial

Modern Physics

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(Dated: May 7, 2026)

For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity. Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK 11

SPECIAL RELATIVITY

- **Increase of momentum with speed / Ex. 2.60**
  A particle initially has a speed of $u_0 = 0.5\,c$ and momentum p_0. At what speed u does its momentum p increase by (a) 1 %, (b) 10 %, (c) 100 %?

  Solution
  The ratio between the final and initial momentum is

  $$\epsilon = \frac{p}{p_0} = \frac{\gamma m u}{\gamma_0 m u_0}$$

  Hence, since u_0 and γ_0 are fixed, solving for u yields

  $$u = \epsilon\gamma_0 u_0 \left[1 + \frac{1}{c^2}\left(\epsilon\gamma_0 u_0\right)^2\right]^{-1/2}.$$

  For (a) $\epsilon = 1.01$ one gets $u = 0.504\,c$, for (b) $\epsilon = 1.1$ one gets $u = 0.536\,c$, for (c) $\epsilon = 2$ one gets $u = 0.756\,c$.

- **Relativistic circular motion / Exs. 2.55 + 2.56**
  Newton's second law is given by $\vec{F} = d\vec{p}/dt$. If the force is always perpendicular to the velocity $\vec{v}$ (as in the case of uniform circular motion), show that $\vec{F} = \gamma m\vec{a}$, where $\vec{a}$ is the acceleration.
  Derive the radius of the circular path of a particle with charge q traveling with speed v in a magnetic field B perpendicular to the particle's path. What happens to the radius as the speed increases, as in a cyclotron?

  Solution
  Using the relativistic form of Newton's second law:

  $$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(\gamma m\vec{v}) = \gamma^3\frac{\vec{v}\cdot\vec{a}}{c^2}m\vec{v} + \gamma m\vec{a},$$

  where we used the explicit form of the relativistic factor $\gamma = \left(1 - \vec{v}^2/c^2\right)^{-1/2}$. Let us multiply both sides with $\vec{v}$, and use that $\vec{v}$ is perpendicular to $\vec{F}$:

  $$0 = \vec{v}\cdot\vec{F} = \gamma^3\frac{\vec{v}\cdot\vec{a}}{c^2}m\vec{v}^2 + \gamma m\vec{v}\cdot\vec{a} = \gamma m\left(1 + \gamma^2\frac{\vec{v}^2}{c^2}\right)\vec{v}\cdot\vec{a} = \gamma^3 m\vec{v}\cdot\vec{a},$$

  from which we see that $\vec{v}\cdot\vec{a} = 0$, the acceleration is perpendicular to the velocity. Substituting this back into the previous equation we find $\vec{F} = \gamma m\vec{a}$. A peculiar consequence of the relativistic Newton's equation, is that the acceleration is in general *not parallel* to the force. The situation when the force is parallel or perpendicular to the velocity is special, it is possible to show that these are the only cases when the acceleration is parallel to the force.

  In the following, we omit the vector for simplicity, and we focus on the modulus. In a circular motion, the centripetal force is given by

  $$F_c = \gamma m a_c = \gamma m\frac{v^2}{r},$$

  and for a charged particle in a magnetic field, the centripetal acceleration keeping the particle in the circular orbit is provided by Lorentz's force

  $$F_L = qvB = F_c = \gamma m\frac{v^2}{r},$$

  yielding

  $$r = \frac{\gamma m v}{qB} = \frac{p}{qB},$$

  showing that for a given magnetic field the radius is increased compared to the non-relativistic case. This is a problem in circular particle accelerators (cyclotrons), where extreme high magnetic fields, or large rings are

needed to reach ultrarelativistic speeds. Note that the result is formally identical to the Newtonian case, with the momentum replaced with its relativistic counterpart $p = \gamma m v$.

We can also compute the *cyclotron frequency*:

$$\omega = \frac{v}{r} = \frac{qB}{\gamma m},$$

showing that for a given magnetic field the cyclotron frequency is reduced compared to the non-relativistic case.

- **Relativistic energy**
  Derive the relativistic energy formula starting from the relativistic momentum $\vec{p} = \gamma m_0 \vec{v}$, the equation of motion $\vec{F} = \frac{d\vec{p}}{dt}$ and using that the work done on a particle is equal to the change of its energy:

$$E_2 - E_1 = W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}.$$

Solution
Substituting into the formula, treating everything as a function of time, and changing the integration variable to t:

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r} = \int_{t_1}^{t_2} \frac{d\vec{p}}{dt} \cdot \vec{v}\, dt = \int_{t_1}^{t_2} \frac{d}{dt}(\gamma m_0 \vec{v}) \cdot \vec{v}\, dt.$$

Next we integrate by parts:

$$W = \left[\gamma m_0 \vec{v}^2\right]_{t_1}^{t_2} - \int_{t_1}^{t_2} (\gamma m_0 \vec{v}) \cdot \frac{d\vec{v}}{dt}\, dt = \left[\gamma m_0 \vec{v}^2\right]_{\vec{v}_1}^{\vec{v}_2} - \int_{\vec{v}_1}^{\vec{v}_2} (\gamma m_0 \vec{v}) \cdot d\vec{v},$$

where we changed the integration variable to $\vec{v}$ and introduced $\vec{v}_{1/2} = \vec{v}(t_{1/2})$. We notice that

$$\frac{d}{d\vec{v}}\gamma^{-1} = \frac{d}{d\vec{v}}\left(1 - \frac{\vec{v}^2}{c^2}\right)^{1/2} = -\frac{1}{2}\frac{2\vec{v}}{c^2}\left(1 - \frac{\vec{v}^2}{c^2}\right)^{-1/2} = -\frac{\gamma \vec{v}}{c^2},$$

which gives us the primitive function in the integral:

$$W = \left[\gamma m_0 \vec{v}^2\right]_{\vec{v}_1}^{\vec{v}_2} + \left[\frac{m_0 c^2}{\gamma}\right]_{\vec{v}_1}^{\vec{v}_2} = \left[\gamma m_0 \left(\vec{v}^2 + \frac{m_0 c^2}{\gamma^2}\right)\right]_{\vec{v}_1}^{\vec{v}_2} = \left[\gamma m_0 \left(\vec{v}^2 + c^2\left(1 - \frac{\vec{v}^2}{c^2}\right)\right)\right]_{\vec{v}_1}^{\vec{v}_2} = \left[\gamma m_0 c^2\right]_{\vec{v}_1}^{\vec{v}_2}.$$

The work is equal to change of the function $\gamma m_0 c^2$, which we identify as the energy of the particle

$$E = \gamma m_0 c^2 = \frac{m_0 c^2}{\sqrt{1 - \frac{\vec{v}^2}{c^2}}} = m_r c^2.$$

From this derivation we only get the energy up to an overall constant, the relativistic energy is defined in a way that the rest energy (corresponding to zero velocity) equals $m_0 c^2$.

- **Relativistic momentum and energy / Ex. 2.94** An electron has a total energy $E = 250E_0$, where E_0 is its rest energy. Determine its (a) kinetic energy, (b) speed, and (c) momentum.

  Solution
  The relativistic kinetic energy is given by

  $$K = E - E_0 = 249E_0 = 249m_0c^2 = 127\,\text{MeV},$$

  where the rest energy of the electron (to three significant digits, and making the rounding explicit) is $m_0c^2 = 0.510(9)\,\text{MeV}$.

  One can extract β considering that

  $$E = \gamma E_0 \Rightarrow \gamma = 250,$$

  and

  $$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = 0.999\,992.$$

  Finally, the relativistic momentum is given by

  $$p^2c^2 = E^2 - E_0^2,$$

  so that

  $$p = \frac{\sqrt{E^2 - E_0^2}}{c^2} = \frac{1}{c}\, m_0c^2\sqrt{250^2 - 1} = 128\,\text{MeV}/c.$$

- **Kinetic energy of different particles / Ex. 2.82**
  What is the kinetic energy of (a) an electron having a momentum of $p = 40\,\text{GeV}/c$? (b) a proton having a momentum of $p = 40\,\text{GeV}/c$?

  Solution
  Knowing the momentum and rest energy of each particle, one can calculate the energy

  $$E^2 = p^2c^2 + m_0^2c^4$$

  and then the kinetic energy as

  $$K = E - E_0 = E - m_0c^2.$$

  For the electron, $m_0c^2 = 0.510(9)\,\text{MeV}$ which is negligible with respect to its momentum, so to three significant digits $K = 40.0\,\text{GeV}$.

  Instead, the proton is much *heavier* ($m_p = 0.938(2)\,\text{GeV} \approx 2000\,m_e$) and therefore

  $$K = \sqrt{p^2c^2 + (m_0c^2)^2} - m_0c^2 = 39.0(7)\,\text{GeV}.$$

- **Energy to accelerate a spaceship / Ex. 2.76**
  Calculate the energy needed to accelerate a spaceship of mass $10^4\,\text{kg}$ to a speed of $v = 0.3\,c$ for intergalactic space exploration. Compare this with a projected annual energy usage on Earth of $10^{21}\,\text{J}$.

  Solution
  At cruise speed, the spaceship's kinetic energy is given by

  $$K = (\gamma - 1)mc^2 = \left(\frac{1}{\sqrt{1 - \beta^2}} - 1\right) mc^2 = 4.35 \times 10^{19}\,\text{J},$$

  which is about $4\,\%$ of the annual energy usage on Earth.

- **Compton scattering**
  Derive the Compton scattering formula using four-momentum conservation.

  Solution
  Compton scattering is the process of a photon scattering on an electron (or more generally on some charged particle, but as we will see, the effect is largest for low mass particles).

  We use the four-momentum conservation, for a general relativistic particle the four-momentum is given by

$$P = \begin{pmatrix} \frac{E}{c} \\ p_x \\ p_y \\ p_y \end{pmatrix} = \begin{pmatrix} \sqrt{m_0^2 c^2 + p^2} \\ p_x \\ p_y \\ p_y \end{pmatrix},$$

  where we used the relativistic energy-momentum relation $E = \sqrt{m_0^2 c^4 + p^2 c^2}$ where m_0 is the rest mass. An important special case is for massless particles (e.g. photons) where $m_0 = 0$, and the relation simplifies to

$$P_\gamma = \begin{pmatrix} |\vec{p}| \\ p_x \\ p_y \\ p_y \end{pmatrix}.$$

  We also use the relation between the energy and frequency of a photon $E_\gamma = hf = hc/\lambda = c|\vec{p}|$.

  We consider a setup where initially the electron is at rest and the incoming photon propagates in the x direction, with initial four-momenta

$$P_e^{(i)} = \begin{pmatrix} m_e c \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad P_\gamma^{(i)} = \begin{pmatrix} \frac{hf}{c} \\ \frac{hf}{c} \\ 0 \\ 0 \end{pmatrix}.$$

  After the collision the photon leaves with frequency f' in the xy plane at angle θ compared to the x axis, and the electron with momentum p_e at angle $-\phi$ also in the xy plane, giving the final four-momenta:

$$P_e^{(f)} = \begin{pmatrix} \sqrt{m_e^2 c^2 + p_e^2} \\ p_e \cos\phi \\ -p_e \sin\phi \\ 0 \end{pmatrix}, \qquad P_\gamma^{(f)} = \begin{pmatrix} \frac{hf'}{c} \\ \frac{hf'}{c} \cos\theta \\ \frac{hf'}{c} \sin\theta \\ 0 \end{pmatrix}.$$

  Conservation of the total four-momentum means $P_e^{(i)} + P_\gamma^{(i)} = P_e^{(f)} + P_\gamma^{(f)}$, in components

$$\begin{pmatrix} m_e c + \frac{hf}{c} \\ \frac{hf}{c} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{m_e^2 c^2 + p_e^2} + \frac{hf'}{c} \\ p_e \cos\phi + \frac{hf'}{c} \cos\theta \\ -p_e \sin\phi + \frac{hf'}{c} \sin\theta \\ 0 \end{pmatrix}.$$

  Multiplying all equations by c, rearranging the first equation (energy, or timelike component) and taking its square

$$\begin{aligned} (m_e c^2 + h(f - f'))^2 &= m_e^2 c^4 + p_e^2 c^2 \\ p_e^2 c^2 &= h^2 (f - f')^2 + 2h m_e c^2 (f - f'). \end{aligned}$$

  We also rearrange the x and y component equations to bring terms containing p_e to the right, square and add the two equations:

$$\begin{aligned} h^2 (f - f' \cos\theta)^2 + h^2 f'^2 \sin^2\theta &= p_e^2 c^2 \\ h^2 (f^2 - 2ff' \cos\theta + f'^2) &= p_e^2 c^2 \\ h^2 (f - f')^2 + 2h^2 ff' (1 - \cos\theta) &= p_e^2 c^2. \end{aligned}$$

We substitute the previous expression for $p_e^2c^2$, simplify and divide by ff':

$$\begin{aligned} h^2(f-f')^2 + 2h^2ff'(1-\cos\theta) &= h^2(f-f')^2 + 2hm_ec^2(f-f') \\ hff'(1-\cos\theta) &= m_ec^2(f-f') \\ h(1-\cos\theta) &= m_ec^2\left(\frac{1}{f'}-\frac{1}{f}\right) \\ \frac{1}{f'} &= \frac{1}{f} + \frac{h}{m_ec^2}(1-\cos\theta) \\ \lambda' &= \lambda + \frac{h}{m_ec}(1-\cos\theta), \end{aligned}$$

where we substituted the wavelength $\lambda = c/f$. The quantity appearing on the right hand side is the so called Compton wavelength

$$\lambda_c = \frac{h}{m_ec} \approx 2.426\,\text{pm},$$

which controls the size of the effect. We see that the effect has a high relative magnitude if the photon wavelength is comparable to the Compton wavelength (or put differently, the photon energy is comparable to the electron rest energy), requiring X-ray or gamma photons. The effect is inversely proportional to the mass of the charged particle, that is why it is easiest to observe with electrons.

- **Compton effect at $90°$ / Ex. 3.53**
  A photon of wavelength $\lambda = 2.0\,\text{nm}$ Compton-scatters from an electron at an angle of $\theta = 90°$. What is the modified wavelength and the fractional change, $\Delta\lambda/\lambda$?

  Solution
  The modified wavelength due to Compton scattering is given by

  $$\lambda' = \lambda + \frac{h}{mc}\left[1-\cos(\theta)\right],$$

  for $\theta = 0$ the wavelength is modified by the *Compton wavelength*

  $$\lambda_c = \frac{h}{mc} = 2.426\times10^{-3}\,\text{nm}.$$

  Alternatively one can use $hc = 1240\,\text{eV nm}$ and the electron rest energy $mc^2 = 0.511\,\text{MeV}$ and in any case get

  $$\lambda' = 2.002\,43\,\text{nm},$$

  where we used six significant digits since the relative variation is small

  $$\frac{\Delta\lambda}{\lambda} = 1.21\times10^{-3}.$$

- **Doppler effect: traffic light / Ex. 2.48**
  An astronaut is said to have tried to get out of a traffic violation for running a red light (wavelength $\lambda_r = 650\,\text{nm}$) by telling the judge that the light appeared green (wavelength $\lambda_g = 540\,\text{nm}$) to her as she passed by in her high-powered transport. If this is true, how fast was the astronaut going?

  Solution
  In the relativistic Doppler effect, the frequency f perceived by an observer is modified with respect to the frequency emitted by the source f_0, depending on the speed of the observer relative to the source

  $$f = f_0\sqrt{\frac{1+\beta}{1-\beta}},$$

  where $\beta > 0$ if the the observer and the source are approaching and $\beta < 0$ is they are receding. Since the wavelength is $\lambda = c/f$, then

  $$\lambda = \lambda_0\sqrt{\frac{1-\beta}{1+\beta}}.$$

  Solving for β yields $\beta = 0.183$ or $v = 0.183\,c = 0.549\times10^8\,\text{m/s}$. In addition to a red light violation, the astronaut gets a speeding ticket.

- **Line spectra of H emitted by a moving star / Ex. 3.14**
  Suppose that a detector in a new space telescope will be capable of detecting visible light in the wavelength range of $\lambda_{\min} = 400\,\text{nm}$ to $\lambda_{\max} = 700\,\text{nm}$. (a) List all the wavelengths for the hydrogen atom that are in this range and their series name. (b) The detector measures visible wavelengths of $\lambda_1 = 537.5\,\text{nm}$, $\lambda_2 = 480.1\,\text{nm}$, and $\lambda_3 = 453.4\,\text{nm}$, that researchers believe are due to the hydrogen atom. Why are all the known visible hydrogen lines not detected? (c) Use these data to calculate the speed of the stellar object that emitted the spectra. Assume that the object is not rotating. Why might rotation be an issue?

  Solution
  The wavelengths of the hydrogen atom in the visible range are the first four of the Balmer series ($n = 2$) given by

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$$

  for $n = 2$ and $n < k \leq 6$ one indeed finds

  - $k = 3$: $\lambda_{2\to3} = 656.5\,\text{nm}$ (red)
  - $k = 4$: $\lambda_{2\to4} = 486.3\,\text{nm}$ (blue-green)
  - $k = 5$: $\lambda_{2\to5} = 434.2\,\text{nm}$ (violet)
  - $k = 6$: $\lambda_{2\to6} = 410.3\,\text{nm}$ (violet)

  while for $k = 7$ the next line is at $\lambda_{2\to7} = 397.1\,\text{nm} < 400\,\text{nm}$. The wavelengths of the Lyman series ($n = 1$) are in the ultraviolet region and those of the Paschen series ($n = 3$) are in the infrared region.

  (b) In principle, the $k = 6$ line of hydrogen should also fall within the visible range. This means that the source emitting the light is moving. One can find the velocity of the source by calculating the *redshift* of the experimentally observed lines with respect to their theoretical values. We can identify the observed lines by calculating the ratios of the wavelengths and comparing them to the ratios of the reference lines of hydrogen at rest. We find that $\lambda_1/\lambda_2 \approx 1.12$ which coincides with the ratio of the $2 \to 4$ and $2 \to 5$ Balmer transitions' lines, the ratio is $\lambda_{2\to4}/\lambda_{2\to5} = \frac{28}{25} = 1.12$. Indeed the observed wavelengths are longer than the theoretical ones, meaning that the corresponding frequency is lower and that the source is therefore moving away from the observer. From this we find that the Doppler factor is $\lambda_{\text{observed}}/\lambda_{\text{source}} = \lambda_1/\lambda_{2\to4} \approx 1.105$. The same ratio is obtained with any other pair of observed and reference wavelengths. This also explains why the $\lambda_{2\to3}$ red line is not visible, its wavelength is pushed out of the visible range. However, the $\lambda_{2\to7}$ ultraviolet line (and possibly further lines) should be visible to the detector, as its wavelength is pushed into the visible range at 438.9 nm, this is an error in the problem.

  Using the formula of the relativistic Doppler effect for the wavelength

$$\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} = \sqrt{\frac{1-\beta}{1+\beta}}.$$

  Solving for beta one gets

$$\beta = \frac{1 - \left(\dfrac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} \right)^2}{1 + \left(\dfrac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} \right)^2}.$$

  calculating this for, e.g., $k = 3$, one obtains $\beta = -0.0995 \approx -0.10$ or $v \approx -0.10\,c = -3.0 \times 10^7\,\text{m/s}$, where the mius sign indicates that the source is moving away from the observer.

  If the object were rotating, one side would move towards the observer and the other away from the observer. This would result in a range of wavelengths observed by the detector.

- **Kaon decay / Ex. 2.103**
  One possible decay mode of the neutral kaon is $K^0 \to \pi^0 + \pi^0$. The corresponding rest energies of the particles are $E_{0\,K^0} = 498\,\mathrm{MeV}$ and $E_{0\,\pi^0} = 135\,\mathrm{MeV}$. The kaon is initially at rest when it decays. (a) How much energy is released in the decay? (b) What are the momentum and relative directions of the two neutral pions (π^0)?

  Solution
  (a) In the frame of the decaying K_0 meson, the initial energy is

  $$E = E_{0\,K^0} = 498\,\mathrm{MeV}$$

  and since the energy is conserved in the decay process

  $$E = K + 2E_{0\,\pi^0},$$

  so that the difference of rest masses is transformed into kinetic energy available for the π^0 mesons

  $$K = E_{0\,K^0} - 2\,E_{0\,\pi^0} = 228\,\mathrm{MeV}$$

  In that reference frame, the π^0 mesons must recoil with equal momenta in opposite directions in order to conserve momentum. Hence, the kinetic energy of each π^0 meson is

  $$K_{\pi^0} = K/2 = 114\,\mathrm{MeV},$$

  and the corresponding energy is

  $$E_{\pi^0} = K_{\pi^0} + E_{0\,\pi^0} = 249\,\mathrm{MeV},$$

  Finally the (modulus of the) momentum can be obtained as

  $$p = \frac{\sqrt{E^2 - E_0^2}}{c} = 209\,\mathrm{MeV}/c.$$